Friedrich-Alexander-Universität
Department Artificial Intelligence
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IDEA Lab
Image Data Exploration
and Analysis Lab

Extending Geometric Transformations to Tabular Data

Idea Lab Report

Seminar Paper

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Image Data Exploration and Analysis (IDEA) Lab

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Abstract

This report documents the reproduction and extension of the self-supervised anomaly detection method originally proposed by Golan and El-Yaniv (2018). While the original work focuses on geometric transformations for images, this project extends the paradigm to tabular financial data. We introduce a *TabularTransformer* and specialized architectures (Robust ResNet, Wide MLP) to detect fraud in high-dimensional datasets. Our experiments on Credit Card and Elliptic datasets demonstrate that transformation-based learning significantly outperforms classical baselines like Isolation Forests in complex fraud detection scenarios.

0.1 Methodological Extensions

GitHub Repository: https://github.com/Andre1917/AnomalyDetectionTransformations

The core contribution of this project is the adaptation of the geometric transformation method from the image domain to tabular data. Since geometric operations (e.g., rotation, flipping) are undefined for tabular features, I implemented a new **TabularTransformer** module. Instead of geometric shifts, this module applies feature-space corruptions to create self-supervised tasks. The transformations defined include: (1) Identity, (2) Random noise injection (Gaussian), (3) Zero-masking of feature subsets, and (4) Permutation of column pairs.

Furthermore, the model architecture was extended to handle non-spatial data, replacing the original simple CNN with architectures tailored for tabular inputs:

- **Robust ResNet:** For the *CreditCard* and *Elliptic* datasets, a Residual Network with skip-connections and batch normalization was implemented. This addresses the vanishing gradient problem and preserves the signal of subtle anomalies.
- **Wide MLP:** For the high-dimensional *IEEE-CIS* dataset, a wide Multilayer Perceptron was implemented to capture complex feature interactions without spatial bias.

Technically, the codebase was modernized to run on a Dockerized environment, resolving compatibility issues with legacy Python/TensorFlow versions.

0.2 Experiments and Results

Experiments were conducted on three datasets: *Credit Card Fraud Detection*, *Elliptic Bitcoin Dataset*, and *IEEE-CIS Fraud Detection*. The primary metric used is the Area Under the Receiver Operating Characteristic curve (ROC-AUC).

0.2.1 Comparison with Baselines

We compared the proposed transformation-based method against classical benchmarks: Isolation Forest (IF) and Autoencoders (AE). As shown in Table 1, our approach significantly outperforms the Isolation Forest baseline on the Credit Card dataset (+0.497 AUC) and the Elliptic dataset (+0.514 AUC).

Table 1 ROC-AUC comparison between Baselines and the Extended Transformation Method (Ours).

Dataset	Isolation Forest	Autoencoder	Ours (New Results)
Credit Card Fraud	0.446	0.844	**0.943**
Elliptic Bitcoin	0.129	0,528	**0.747 (NoAggregated)**
IEEE-CIS Fraud	**0.635**	0.565	0.620

On the *Credit Card* dataset, the Robust ResNet successfully learned to distinguish fraud by identifying which "puzzle" (transformation) was applied to the transaction features, achieving a state-of-the-art score of 0.943. The classical Isolation Forest failed to capture the complex fraud patterns, likely due to high dimensionality and class imbalance.

0.2.2 Ablation Study: Feature Importance in Elliptic

For the *Elliptic Bitcoin* dataset, an ablation study was conducted to analyze the impact of feature groups. The results revealed a counter-intuitive finding:

- **Baseline (All Features):** AUC 0.511
- **No Aggregated Features (Only Local):** AUC 0.643

Removing the aggregated graph features significantly improved performance. This suggests that pre-calculated aggregates introduced noise that confused the self-supervised task, whereas raw local features (fees, timestamps) contained the most relevant signals.

Declaration of Academic Integrity

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Hallerndorf, 28th April 2026

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